VUA-SEN307 DATABASE SYSTEMS — WEEK 4 PRACTICAL

Lab 2: Database Design and Normalization | Name: NTAH CHIDERA ISAAC | Matric No.: VUG-SEN-24-11663

Unnormalized registration table (UNF)

Matric No.	Student Name	Student Dept.	Course Codes	Course Titles	Lecturers	Lecturer Depts.	Session
20231234	Ada Obi	Software Engineering	SEN307, SEN301	Database Systems, Software Architecture	Dr A. Okafor, Dr M. Bello	Software Eng., Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307, COS301	Database Systems, Operating Systems	Dr A. Okafor, Mrs F. Eze	Software Eng., Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

First Normal Form (1NF) — PK: (matric_no, course_code, session)

Matric No.	Student Name	Student Dept.	Course	Course Title	Lecturer	Lecturer Dept.	Session
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.	2026/2027
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Computer Sci.	2026/2027
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.	2026/2027

Explanation: Removed multi-valued cells and gave each student-course combination its own row so each cell contains one value.

Second Normal Form (2NF)

matric_no (PK)	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

course_code (PK)	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

matric_no (PK/FK)	course_code (PK/FK)	session (PK)
20231234	SEN307	2026/2027
20231234	SEN301	2026/2027
20231235	SEN307	2026/2027
20231235	COS301	2026/2027
20231236	SEN301	2026/2027

Explanation: Separated student and course details from enrolment records to remove partial dependencies on the composite key.

Third Normal Form (3NF) — Final schema

Department (dept_id **PK**, dept_name)
Student (student_id **PK**, matric_no **UNIQUE**, full_name, dept_id **FK** → Department.dept_id)
Lecturer (lecturer_id **PK**, lecturer_name, dept_id **FK** → Department.dept_id)
Course (course_id **PK**, course_code **UNIQUE**, course_title)
Enrolment (student_id **PK/FK** → Student, course_id **PK/FK** → Course, session **PK**)
TeachingAssignment (lecturer_id **PK/FK** → Lecturer, course_id **PK/FK** → Course, session **PK**)

Example data: Department: (1, Software Engineering), (2, Computer Science). Lecturers: (1, Dr A. Okafor, dept 1), (2, Dr M. Bello, dept 1), (3, Mrs F. Eze, dept 2). TeachingAssignment for 2026/2027: (lecturer 1, SEN307), (lecturer 2, SEN301), (lecturer 3, COS301).

Explanation: Moved department details into Department, linked students and lecturers to departments with foreign keys, and used TeachingAssignment to record which lecturer teaches which course in each session.